MS0003: Lab Quiz 2 (100 marks)

- Submit a SINGLE Jupyter Notebook file named **StudentID.ipynb**, where StudentID is your matriculation number.
- Download the data file (in CSV format) posted corresponding to this Lab Quiz. You will need it for the problems.
- If you are supposed to "Comment" something extra, use a Markdown cell in the notebook or include standard Python #comments in the cells.

Problem 1: Classification (35)

Context

`nutrition_dataset_simp.csv`: The dataset contains nutritional information about different products that a supermarket sells: iron, carbohydrate, fiber, fat and water. Your goal is to connect the variables and predict whether the food is vegetarian or not (vegetarian/non- vegetarian) using the other variables in the dataset, using a Decision Tree.

a) Print the overall statistical description of the `nutrition_dataset_simp.csv` data and plot standard statistical distributions for each variable, including for the categorical variable. **(6 marks)**

b) Calculate the ratio of "non-vegetarian/vegetarian" in category. Print this ratio. **(3 marks)**

c) Partition the dataset <u>randomly</u> into Train and Test datasets: 73% for Train and 27% for Test. On the Train set, fit **a multi-variate 4-level** Decision Tree model for *category* against all six other numeric variables. **(6 marks)**

d) Predict *category* using this model on both Train and Test datasets. Print the Classification Accuracy in both cases. **(3 marks)**

e) Compute from the Confusion Matrix, the values of false positive rate (fpr) and the false negative rate (fnr) and print these values. **(5 marks)**

f) Partition the dataset **<u>again</u>** into new Train and Test datasets: 73% for Train and 27% for Test and set `shuffle = False` within the arguments of the train_test_split function. On this Train set, fit **a multi-variate 3-level** Decision Tree model for *category*, **<u>plot the decision tree and comment on which model is better and why</u>**. ***Note:*** `shuffle = False` makes sure the split is not random. **(7 marks)**

g) Create a new dataframe with the entries below. Based on your model in 2(d), predict the category for each of the combinations and use the print function to display the category prediction (Vegetarian/Non-vegetarian). **(5 marks)**

Iron	carbohydrate	fiber	fat	water	calories	category
3.19	0	0	8.62	65.09	179	
66.36	44.24	10.5	22.27	8.06	375	
0.23	5.88	3	0.18	92.3	25	
2.15	47	2.6	20.9	21	414	
0.37	0.74	0	0.97	79.18	86	

(continued on next page)

Problem 2: Clustering (35)

Context

`Pupil.csv`: This dataset contains 150 instances that can be differentiated based on 4 attributes.

a) Load `Pupil.csv`. Visualize the data and conduct statistical analysis. Comment on the distribution. **(12 marks)**

b) Perform K-means clustering and K-Means++. For both clustering methods, draw an elbow plot (y-axis: within sum of squares, x-axis: number of clusters) and decide the optimal number of clusters. Justify your choice. **(14 marks)**

c) Interpret the clusters. (**5 marks)**

d) Perform anomaly detection. Use 0.1**A**%, where **A** is the **last digit of your student ID** as contamination fraction. Use (10+**A**) as the number of neighbours. (**4 marks)**

Problem 3: Neural Networks (30)

Context

The ***Iris dataset*** represents different types of Iris flowers. Each Iris type can be differentiated based on its petal length, petal width, sepal length and sepal width, as per Figure 1 below. This dataset is one of the earliest used in classification literature. It was originally presented by R.A. Fisher in his 1936 paper "The use of Multiple Measurements in Taxonomic Problems"

a) Load the ***Iris dataset*** from scikit-learn library and split it into data and target. Create a classification MLP to classify types of flowers using all variables. Use an 80-20 train/test split with random_state = 15. Generate a classification score and confusion matrix for both train and test dataset. Parameters for MLP: `max_iter=30`. **(15 marks)**

b) Generate a pair-plot of all variables, colored by flower type classification. By visually analyzing the pair plot, comment on which features is the most dominating in classifying the flower types. Justify your answer. **(8 marks)**

c) Conduct a grid search optimization MLPClassifier (`max_iter = 100`) hyperparameters. Use the following parameter space:

```
parameter_space = {
    'hidden_layer_sizes': [(10,20,10),(5,10,5)],
    'activation': ['tanh', 'relu'],
    'solver': ['sgd', 'adam'],
    'alpha': [0.0001, 0.0002],
    'learning_rate': ['constant','adaptive'],
}
```

Print out the best parameters chosen by the grid search and comment on the effect of each parameter chosen. Print out the classification accuracy of train and test model. **(7 marks)**